

Dr. D. Y. Patil Unitech Society's
Dr. D. Y. Patil Arts, Commerce and Science College, Pimpri, Pune
Academic Year 2024-25
F.Y.B. Sc.(Data Science) (Sem- I)
Subject: Computer Organization
QUESTION BANK

SHORT ANSWERS (1 or 2 Marks)

Unit 1

1. Define number of system.
2. Define binary number system.
3. Define Hexadecimal number system.
4. Define base
5. Define gray code
6. Define ASCII
7. Define Logic gate
8. State De morgan's 1st & 2nd Theorem.
9. What is K-map?
10. What is base for binary & hexadecimal.
11. List no of variable eliminated in pair ,quad & octal in k- map
12. Define multiplexer.
13. Define demultiplexer.
14. Symbol for all gates
15. Why name Universal gate ?
16. Define encoder
17. Define decoder
18. Define flip-flop
19. Define counters
20. Define shift register
21. Truth table for T and D flip flop
22. Applications of shift register.
23. Types of shift registers
24. Define high level ,low level, positive edge and negative edge triggering
25. Expression for all logic gates
26. Truth table for all logic gates

Unit 2

27. List functions of CPU.
28. Define PC, SP, IP, IR, ID, Flag, MAR, MBR.
29. Define cache memory
30. Define virtual memory
31. Define segmentation & paging
32. Define interrupt.
33. List types of data transfer methods
34. Define serial communication & its types

Unit 3

35. Define parallelism
36. Define pipelining
37. Define arithmetic pipelining
38. Define RISC
39. Define CISC
40. Define Numeric co processor.
41. List and define function of all the registers used in 8086
42. List and define function of all the flags used in 8086

LONG ANSWER QUESTIONS (3 or 4 marks)

Unit 1

1. Explain binary number of system.
2. Explain Hexadecimal number of system.
3. Explain the conversion from binary to hexadecimal with example.
4. Explain the conversion from hexadecimal to binary with example.
5. Explain gray code with table
6. Explain BCD code in detail.
7. Explain Classification of logic gates
8. Explain the all the gates with truth table, expression and symbol.
9. Simplification problems of logic units
10. Write short note on ASCII
11. State and prove De-Morgan's 1st and 2nd theorem
12. Explain the structure of 2, 3 and 4 variable K map
13. Solve all the problems based on K map
14. Explain Half adder with logic diagram
15. Explain full adder with logic diagram
16. Explain half subtractor with diagram

17. Explain 2:1 and 4:1 multiplexer
18. Explain 1:2 and 1:4 demultiplexer
19. Explain decimal to BCD encoder with diagram
20. Explain 3 to 8 decoder with diagram
21. Explain SR Flip flop with diagram
22. Explain JK flip flop with diagram
23. Explain D and T flip flop with diagram
24. Differentiate between synchronous and asynchronous counter.
25. Define shift registers, its types and applications
26. Explain SISO, SIPO, PISO, PIPO shift registers with diagrams.

Unit 2

27. Explain the functions of CPU
28. Explain the general register organization in CPU
29. Explain the general registers used in CPU.
30. Explain the concept of Stack
31. Explain memory hierarchy
32. Explain cache memory
33. Explain associative memory and virtual memory
34. Differentiate between segmentation and paging
35. Explain the block diagram of parallel interfacing
36. Explain the concept of interrupt
37. Explain the types of input/output data transfer
38. Explain R-2R DAC
39. Explain ADC.
40. Explain serial communication and its types

Unit 3

41. Explain the block diagram of 8086 in detail
42. Explain 8086 registers
43. Explain numeric co processor with block diagram
44. Explain the concept of parallelism and pipelining
45. Explain instruction and arithmetic pipelining
46. Differentiate between RISC and CISC.
